Reading the Indus Seals: A Translation Corpus of 7,138 Inscriptions

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Overview

This report presents the first mass translation of Indus Valley seal inscriptions using the 605-sign Proto-Dravidian anchor table (Pierson 2026). All readings are computational proposals, not confirmed translations, and should be treated as hypotheses pending peer review.

Metric	Value
Total inscriptions translated	7,138
Corpora	Holdat (1,669 seals) + Yajnadevam (5,469 inscriptions)
Mean sign coverage	87.9%
Fully translated (100% coverage)	5,372 (75.3%)
Archaeological sites represented	77
Unique inscription formulas	3,352
PDr vs Sanskrit side-by-side	3,731 comparisons

The Inscriptions: What Do They Say?

The most common seal readings confirm the guild-identity model proposed in the preprint. Seals encode short formulaic titles — not commodity tallies, not religious texts, but professional identity markers analogous to medieval guild marks.

Most Common Seal Formulas

Reading	Count	Interpretation
kur	284 (4.0%)	"Hill/short" — single-sign administrative marker
min/min	150 (2.1%)	"Fish/star" — fish guild or astronomical marker
kur-kur	136 (1.9%)	Doubled hill marker — emphatic or plural
kur-puli	130 (1.8%)	"Hill-tiger" — tiger-clan hill guild
kur-tol	112 (1.6%)	"Hill-shoulder" — craft guild title

Reading	Count	Interpretation
kur-koL	96 (1.3%)	"Hill-vessel" — jar/pottery guild
koL	60 (0.8%)	"Vessel/jar" — single-sign potter marker
puli	41 (0.6%)	"Tiger" — tiger clan
nallavar	40 (0.6%)	"Good people" — elite/noble title
al-kur-kat	39 (0.5%)	"Man-hill-bond" — compound administrative title

The dominant pattern is **[ANIMAL/CRAFT]-[PLACE/TITLE]-[SUFFIX]** — exactly the tripartite INITIAL-MEDIAL-TERMINAL grammar confirmed at 6.3x above null across 76 sites.

The Indus Valley Civilization: What the Seals Reveal

A Civilization of Guilds

The translated inscriptions paint a picture of a highly organized guild-based society. The IVC (ca. 2600-1900 BCE) was not merely a collection of farming settlements — it was an urban civilization with professional specialization, standardized administrative practices, and a continent-spanning trade network.

Key cultural insights from the translation corpus:

1. **Professional identity seals.** The overwhelming majority of inscriptions encode guild membership and personal identity, not commodity counts or religious dedications. A typical seal reads: "the tiger-clan hill craftsman" or "the vessel-guild man." This is directly analogous to later South Asian guild-seal traditions documented in the Arthasastra (4th century BCE).
2. **Animal-clan system.** The most frequent INITIAL signs are animal names: puli (tiger), erutu (bull), kur (hill-creature), min (fish). Each animal appears to designate a professional clan or guild — the "tiger clan," the "bull guild," the "fish-star guild." This totem-based clan system is consistent with Dravidian kinship structures attested in historical Tamil Sangam literature.
3. **Standardized across 77 sites.** The same formulaic patterns appear at Harappa, Mohenjo-daro, Dholavira, Lothal, and 73 other sites spanning from the Afghan border to Gujarat. This uniformity implies a centralized administrative system with shared scribal conventions — remarkable for a civilization covering 1.5 million square kilometers.

Sites and Their Inscriptions

Site	Inscriptions	Region
Harappa	2,582	Punjab (Pakistan)
Mohenjo-daro	1,896	Sindh (Pakistan)
Dholavira	230	Gujarat (India)
Lothal	200	Gujarat (India)
Kalibangan	195	Rajasthan (India)
Chanhu-daro	73	Sindh (Pakistan)
Nausharo	38	Balochistan (Pakistan)
Banawali	26	Haryana (India)
Lakhanjo-daro	14	Sindh (Pakistan)
Rakhigarhi	13	Haryana (India)

The two great cities — Harappa and Mohenjo-daro — account for 63% of all inscriptions. Dholavira, the third-largest site, is notable for its monumental signboard (the only known large-scale Indus inscription) and its distinctive seal repertoire.

Materials: The Technology of Seals

Material	Count	Notes
Steatite	2,549	Primary seal material — soft stone, easy to carve
Clay	913	Sealings (impressions) and tablets
Faience	596	Glazed ceramic — luxury goods markers
Copper	261	Copper tablets — administrative records
Terracotta	76	Fired clay — permanent records
Ivory	36	Rare luxury — high-status seals

Steatite dominates (46% of all inscribed objects), consistent with its use as the standard seal material throughout the IVC. The presence of 261 copper tablets is significant — copper was expensive, suggesting these were official administrative records rather than personal seals.

Chronological Distribution

The corpus spans the full IVC timeline:

- **Early period**: 49 inscriptions — the script's formative phase
- **Intermediate/Period 3**: 1,392 inscriptions — the mature Harappan peak
- **Late period**: 327 inscriptions — the decline phase before ~1900 BCE

The same formulaic patterns persist across all periods, suggesting remarkable scribal conservatism over 700+ years.

Proto-Dravidian vs Sanskrit: Side-by-Side

The translation corpus includes 3,731 inscriptions where both our Proto-Dravidian reading and Yajnadevam's Sanskrit reading are available. As reported in the preprint, agreement is 0/34 for directly comparable signs — the two hypotheses produce completely different readings for the same inscriptions.

This is not merely a disagreement about phonetic values — the entire semantic framework differs. Our readings produce guild titles and professional identifiers; the Sanskrit hypothesis produces religious invocations and deity names. The archaeological context (commercial seals, administrative tablets, port-city trade goods) strongly favors the guild-identity interpretation.

Limitations

1. **Coverage gap.** 12.1% of sign tokens remain untranslated (66 Yajnadevam glyph IDs with no Mahadevan equivalent).
2. **No confirmed translations.** All readings are computational proposals. The 605-sign anchor table has not been independently validated by epigraphists.
3. **Single-hypothesis.** All readings assume Proto-Dravidian. A formal comparison against Proto-Munda readings would strengthen or weaken this interpretation.
4. **No bilingual text.** Until a bilingual inscription is found, all translations remain probabilistic.

Data Availability

All translation data is open source:

- Full corpus: `outputs/seal_translations.json` and `.csv`
- Cultural summary: `outputs/seal_translations_summary.json`
- PDr vs Sanskrit: `outputs/pdr_vs_sanskrit.csv`
- Anchor table: `backend/reports/INDUS_FINAL_ANCHORS.json`
- Repository: https://github.com/BitConcepts/glossa-lab
- Preprint DOI: https://zenodo.org/records/20381178

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